

Progress report 2 of 10

RAG-Augmented Multi-Agent LLM Framework for Explainable Software Vulnerability Detection Using CWE Knowledge Bases

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A small practice set

This report adds a 12-file Java set and a first simple check. It continues report 1.

What we did

We wrote 12 short Java files for the six weaknesses in report 1. Six files are unsafe. Six are safe.

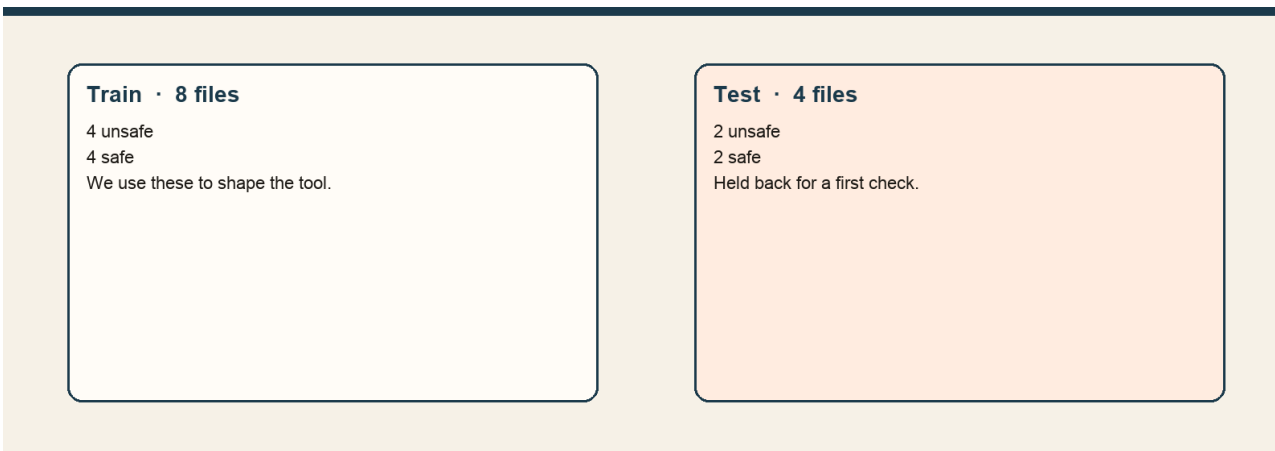
Eight files are for practice. Four files are held back as a first check (two unsafe, two safe).

A simple checker looks for risky text patterns. On these 12 files it found every unsafe file and did not flag a safe file.

F1 is one score that mixes two ideas. Precision is how many alarms were real bugs. Recall is how many real bugs we caught. Here both are 1.000, so F1 is 1.000.

The checker is only a list of text patterns. It does not understand the program. A perfect score on 12 files we wrote ourselves is a first check, not a real test.

Picture



Numbers for this step only

Split	Files	Unsafe	Safe
Train	8	4	4
Test	4	2	2
All	12	6	6

What is next

Next we will store short CWE facts so the tool can look up a name and a fix.

What this report does not claim

A perfect score here does not mean the checker works on real projects. This is not Juliet and not OWASP.